

Automated Oral Cancer Screening from Clinical Photographs Using EfficientNet-Based Transfer Learning

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Abstract—Oral squamous cell carcinoma (OSCC) can be considered one of the most significant health issues faced by the world because the number of late-diagnoses is extremely large. This paper presents a proposal of an automated screening system based on the deep learning approach on clinical intraoral imagery, but with special attention to the effects of model scaling on detection rates. Based on a multi-source dataset of 2,314 clinical images, we tested the EfficientNet family (B0, B1 and B2) to find the best trade-off between the analytical depth and real-life generalization.

In our experiments, we find that there is a non-linear association between the model complexity and the diagnostic accuracy. Although EfficientNet-B0 with its baseline score produced a test accuracy of about 85%, an increase to EfficientNet-B1 produced a colossal performance jump, hitting almost 99% test accuracy and more importantly, dropping the cases of critical false negatives by 14 to only 4. Interestingly, additional scaling to EfficientNet-B2 did not cause significant degradation, indicating that there is some point of diminishing returns, beyond which a large number of parameters will cause overfitting on more narrow medical data.

The models continued to be highly discriminative with values of ROC-AUC exceeding 0.99 across the board. One of the important lessons that can be learned in this work is that moderate scaling offers the best trade-off between sensitivity and generalizability. In order to make sure these systems are not viewed as black boxes, we used Grad-CAM visualizations, which proved that the models offer the option of highlighting clinically relevant lesion features always. Finally, these results highlight that the key to the success of creating fair, implementable medical screening instruments is optimal model selection, as opposed to using maximum complexity as a foundation.

Keywords—Oral cancer detection, oral squamous cell carcinoma (OSCC), deep learning, EfficientNet, clinical image analysis, medical image classification, computer-aided diagnosis, transfer learning, Grad-CAM, explainable AI

I. INTRODUCTION

Oral squamous cell carcinoma (OSCC) is the most common and the most difficult one of head and neck cancers. In spite of the additions to our therapeutic arsenal, the health burden of the world is still high, and thousands of new cases are being diagnosed each year. The main concern is timing, where as the detection in the early stages of the disease has been proven to have a significant impact on the survival after five years, late-stage diagnoses still lead to high mortality.

Nevertheless, it is famously hard to diagnose oral cancer at an early stage. A lot of the potentially malignant lesion are deceptive and quietly resemble benign conditions when it comes to the routine exams. The problem of the lack of specialized treatment is complicated by the absence of this in the world, and the subjectivity of visual examination whereby accuracy may entirely depend on the personal experience of a clinician.

To fill these gaps, scholars have resorted to artificial intelligence (AI), in this case, deep learning, to complement clinical knowledge. Convolutional Neural Networks (CNNs) have already been shown to be of value in medical imaging to recognize high-level patterns that the eye can fail to detect. With regard to oral cancer, the models present a means of standardizing the screening and raising the red flag of subtle signs of malignancy that would have otherwise remained unnoticed.

Although the benchmarks reached by the previous studies are impressive (in many cases, it is in the range of 0.97 and more) they often utilize highly controlled datasets. These laboratory conditions scarcely represent the chaotic variability of the actual clinical images and that can constrain the ability of a model to work in high-stakes hospital setting.

In this paper, we suggest a stronger method of automated screening with EfficientNet-B0 architecture. Transfer learning on a multi-source (2314 clinical images) and diverse dataset has enabled us to create a system with the ability to generalize beyond imaging conditions. To make sure this is not a black box tool we added the Grad-CAM visualizations that would enable clinicians to visualize what the model is specifically highlighting. We aspire to go beyond the mere accuracy and create a system that has clinical trust due to interpretability.

In addition to the point of reference EfficientNet-B0, we rolled out the B1 and B2 models in order to determine the best "capacity threshold" of detecting oral cancer. The hypothesis that is explored in this analysis is that enhancing the depth of a model can enhance generalization in varied clinical contexts or produce overfitting on specialized medical data. Coming up with the comparative of these scales, we hope to determine the architectural sweet spot which provides maximum sensitivity, yet robust performance on undetected real-world images.

The rest of the paper will be structured in the following way: Section II will discuss the existing literature as regards oral lesion detection, whereas in Section III a comparative analysis of the existing methods will be offered. In Section IV and V,

our dataset and methodology are described. Section VI then gives our experimental results and a discussion of future directions is given in Section VII.

II. LITERATURE REVIEW

A paradigm shift in the medical image analysis has been observed over the last decade, and one of the reasons is the advent of deep learning. Convolutional Neural Networks (CNNs) have left pattern matching on simple data to visualisation of complex hierarchical representations of raw data. In some instances, such change is important especially when there is oral cancer screening where the difference between a harmless lesion and the cancerous lesion is mathematically infinitely thin.

The early diagnostic models were very dependent on handcrafted features, i.e. manually defined features, combined with classical algorithms such as SVMs or Random Forests. Although they opened the way, they were necessarily inflexible, and they tended to fail in generalization in variety of real-world clinical data. This has been resolved with the introduction of deep learning, which has made it possible to end-to-end feature learning, meaning of a model scale and adapt to the intricacies of medical imagery without human-imposed restrictions.

The potential is reflected in current research. Multi-stage frameworks are currently used in many studies that first identify a lesion and then categorize it. Although they are highly accurate, they tend to be highly computationally burdened and therefore may not be applicable in real-time clinical activities. Likewise, even though some architectures such as DenseNet and EfficientNet have reported almost perfect AUC figures (often over 0.97), they are often trained under idealized conditions in the lab, i.e. standardized datasets which do not necessarily reflect a busy clinic with poor lighting or camera angles.

Although deep learning has long been more effective than human professionals in particular cases, there is still a problem of trust gap. There are several models which have a limited diversity in data set, class imbalance or lack of interpretability. To trust an AI, a clinician must be aware of the reasons behind a decision. This is the reason why EfficientNet-style architectures have become popular; it employs a compound scaling methodology that trades-off depth and width to achieve high accuracy without the efficiency required to operate in resource constrained settings.

Transfer learning has become a critical approach in environments in which there are small, annotated medical cases, which are uncommon. With this, we can, given a model that has already been trained on large datasets such as ImageNet, fine-tune the network to understand the particulars of the oral pathology. In addition, we have incorporated Grad-CAM (Gradient-weighted Class Activation Mapping) to leave AI out of the black box box. This enables the system to indicate the precise areas of a lesion that raised a flag of malignancy thus making the model examine the pathology and not the clinical noise.

Our research aims at closing the gap between clinical reality and the high-performance theory. Instead of creating a more complicated, multi-stage pipeline, which would need lesion localization manually, we suggest a simplified, one-stage network, which is an EfficientNet-B0. Direct processing of

clinical photographs helps to reduce the computational load and allows much faster inference.

More importantly, we are sensitivity-based in our evaluation. False negative is much more terrible in cancer screening compared to false positive. Training on a multi-source dataset of 2,314 images, we have given a high priority of selecting a robust, interpretable and above all, of a clinically safe model. We are not only aiming at getting a high score on a test set, but it is a tool that can be scaled and a tool that the clinicians can actually believe in the field.

III. COMPARATIVE ANALYSIS

It has been demonstrated that deep learning models, especially CNNs with transfer learning, can reach close to perfect accuracy (97%-99%) when limited to histopathological data. Although such results are technically amazing, they tend to be an ideal work of performance under the condition of the laboratory. The real problem is when attempts are made to transfer scanned slides to clinical reality, i.e. smartphone-taken images or intraoral photographs. Performance in such realistic contexts can easily plummet; smartphone-based systems have been reported to have sensitivities of as low as 83% and some object detectors on raw clinical images have achieved a mean Average Precision (mAP) of as low as 25%. This mismatch is an indication that the models that are born in controlled situations are more likely to collapse in generalization since the physical exam is unpredictable.

This performance drop in the clinic is hardly an inability of the math to function, but an effect of environmental noise. The variations of lighting, angles and bare anatomical diversity of the mouth cavity may present some kind of data shift that is not programmed in most models. The gap has been filled by other researchers who employed intensive preprocessing - normalization, lesion centering, and the resolution standardizing. As much as these measures offer the middle ground on the gains, they introduce the complexity of the system that can serve as a reverse to the very concept of such tools which is supposed to provide real-time, scalable and fast screening.

The existing trend in literature is still super-reliance on transfer learning - utilizing structures such as EfficientNet, ResNet, and DenseNet - combined with data augmentation to scale small labels. The emphasis on sensitivity (recall) is another theme that is repeated in these designs. In a cancer screening situation, false negative (miss) is a much more serious error than a false alarm. Nonetheless, this emphasis causes an incessant conflict of sensitivity and precision, a trade-off, which is often recognized, but not often examined with the depth needed to make it clinically applicable.

The black box problem is, perhaps, the biggest challenge to clinical adoption. Although the explainability techniques are available such as Grad-CAM, these techniques are still secondary considerations in most studies instead of being a fundamental element. Clinician must know the reasoning behind the recommendation of an AI in order to be convinced. Clinical buy-in will never be achieved without clear, visual evidence of the model emphasizing on pathological features, not background noise.

Moreover, the literature is at present limited by homogeneity of the datasets. Most of the studies are retrospective based on small or skewed datasets not externally validated. This commonly results in models that incorrectly place as benign

the features that have not encountered sufficient normal range. Larger, multi-source datasets with true heterogeneity of the patient population are clearly in need and urgently needed.

These are the gaps that our research will aim to fill. Instead, our attention is on real-world clinical image analysis with the help of a multi-source and diverse dataset, instead of training a model ideal on a controlled set of data. Our framework is a

simplified, single stage one stage pipeline, not requiring the complexity of multi-stage pipelines, focusing more on speed and deployment capability. Combining the idea of transfer learning with a sensitivity-based evaluation and incorporating Grad-CAM as a method of interpretation, we are going to leave the realm of raw accuracy. We want like to design a system which is mathematically valid, clinically meaningful, and above all, reliable.

TABLE I. SUMMARY OF DEEP LEARNING STUDIES FOR ORAL CANCER DETECTION

Author	Abstract	Methodology	Accuracy	Limitations
[1]	Proposed a deep-learning system to detect oral lesions in clinical images. Used multi-annotator labels to train models.	Image classification using ResNet-101 and object detection via Faster R-CNN; ensemble of annotations for referral.	87.07%	Limited by small initial dataset (mixed lab-clinic images); preliminary results.
[2]	Employed a modified local binary pattern (MLBP) to capture texture from histopathology images of OSCC vs normal oral mucosa.	Feature extraction (MLBP) combined with deep neural network (MLBP-DCNN) and traditional ML (SVM, KNN, Decision Trees) classifiers on histopathology images.	91.36%	Histopathology dataset only.
[3]	Described a smartphone-based deep learning method with a “center-capturing” rule to improve oral lesion imaging.	Captured oral photos centered on lesions, created a dataset of five disease categories, applied HRNet CNN with resampling to reduce variability	84.3%	Retrospective, medium-sized dataset (455 images).
[4]	Introduced an “Aquila and Gorilla” optimizer framework for TL-based CNNs on histopath slides to detect OSCC.	Transfer-learning on 8 CNN backbones (DenseNet201, VGG, MobileNet, etc.) pretrained on ImageNet, fine-tuned with two metaheuristic optimizers (Aquila, Gorilla) on HOC dataset .	99.25%	large dataset (~5k images) but may have labeling biases; lacks clinical image evaluation.
[5]	Evaluated an ML-based object-detection approach on clinical intraoral photos (360 images of OSCC, OPMDs, healthy mucosa).	Used Roboflow’s annotation tools and trained object detection models (multi-class) on clinician-annotated images.	32.9%	Small, imbalanced dataset; variable image resolutions.
[6]	Explored EfficientNetB3 on a public histopathology set (1224 images) to separate normal vs OSCC epithelium.	Custom EfficientNetB3 CNN with extensive data augmentation and regularization on H&E-stained slides.	99%	Possible dataset bias (class imbalance, patient heterogeneity); risk of overfitting; model interpretability and need for clinical validation are noted challenges.
[7]	Proposed a CNN tailored to histopathologic oral mucosa images to classify OSCC vs normal.	Designed an optimized CNN and compared against standard CNNs (VGG16/19, AlexNet, ResNet, MobileNet, Inception) on an Indian histopath dataset.	97.82%	Only reported cross-validation (no external test); dataset limited to histopathology images.
[8]	Studied DL on clinical photos (smartphone) from 1470 patients for automatic OSCC screening.	Retrospective analysis using CNN models on smartphone oral photos.	96%	Retrospective, mixed data sources; and real-time usability not yet tested.

IV. DATASET AND PREPROCESSING

The current research is conducted on the premise of a multi-source database of 2,314 intraoral clinical image de-identified that have been specifically designed to be a representative sample of the messiness of the imaging in the real world. It has 1,269 malignant (OSCC) and 1,045 non-cancerous samples at a ratio that is marginally biased by the

facts of high-stakes screening clinical practices. We undertook a stringent data audit, in order to develop label consistency and data integrity prior to commencing any type of training. Detection and elimination of duplicates of the samples in aggregated sources, a frequent but too frequently

overlooked pitfall in medical imaging research, was one extremely important aspect of the process.

In order to prepare the data to our deep learning pipeline, we created a standardization of all images into 224 x 224 pixel image resolution, and ImageNet statistics normalization to align with the needs of our transfer learning backbone. Nevertheless, we acknowledge that clinical photography is seldom ideal, and we used a large data augmentation policy to imitate the imprecision of in-the-field photography. We used random resized cropping, horizontal flips, minor rotations ($\pm 15^\circ$), and color jittering (adjusting brightness, contrast and saturation) to make the model learn pathological characteristics instead of depending on a particular light condition or camera orientation.

We divided the dataset into training (70%), validation (15%), and test (15%) subsets in a stratified way to ensure that class representation was similar in all the splits. The set of validation was used as a guide to hyperparameter optimization and early termination, whereas the test set was completely isolated so as to obtain an impartial estimate in the end. In order to be scientifically rigorous, we did strict checks on the data to avoid data leakage so that no duplicated or almost duplicated images would be able to shift between partitions.

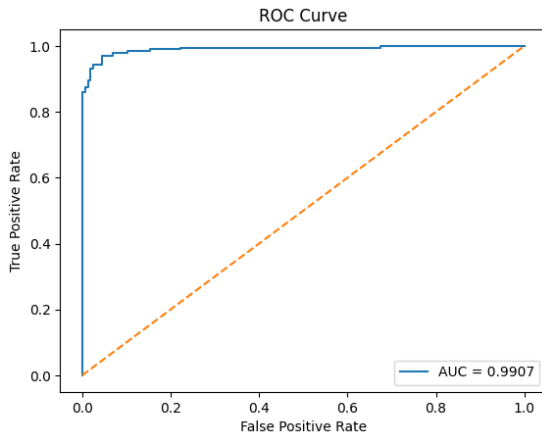


Figure 1: ROC Curve

The resultant model was extremely discriminative with a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.9907. This large measure indicates that the system has a strong capability of differentiating between cancerous and non-cancerous tissues. Although these findings are most promising, we are well aware of the fact that high AUC values under a controlled testing environment are merely but a beginning. To be truly clinically useful, such models need to remain tested in the face of the enormous anatomic and technical heterogeneity that can be found in different populations of patients.

V. METHODOLOGY

We used a strong convolutional neural network classifier that was founded on EfficientNet-B0. EfficientNet-B0 has the least size in the EfficientNet family and a good trade-off between model size and accuracy. It builds upon the concept of compound scaling of network depth, width and image resolution, which it optimizes through an automated neural architecture search (NAS) process. Despite the fact that the original EfficientNet models were created to solve a multi-

class ImageNet classification task, we adjusted the network to our binary screening task.

We began with the EfficientNet-B0, which is pre-trained on ImageNet but replaced the final fully-connected layer (with 1000-way classification) with a two-node output of (malignant, benign). The output was of the form of a SoftMax. We experimented with freezing the initial 80% of layers to preserve pre-trained feature extraction and fine-tuning the remaining few layers, and with fine-tuning all the layers. The practice had shown that the most successful outcome was achieved by fine-tuning all layers with low learning rate, which allowed the network to adapt high-level features to the oral lesion domain. To prevent overfitting, dropout (0.3) was inserted before the final dense layer, and L2 regularization of weights in the fully-connected weight was applied ($l=1e-5$). Figure X (not shown) represents the model architecture and contained about 5.3 million trainable parameters.

In lieu of interpretation, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) to produce heatmaps that show those regions in an image that have the greatest impact on the classification. Grad-CAM Back-propagation of the score of predicted class before SoftMax to the final convolutional layer produces coarse localization maps. Namely, having trained, on any test image we had calculated the Grad-CAM heatmap of the top score in the "malignant" category. Such heatmaps are superimposed on the input images (Figure 2) to visualize the concentration of the network on lesions. Prior researchers have been able to use Grad-CAM to classifiers of oral lesions to perform qualitative validation.

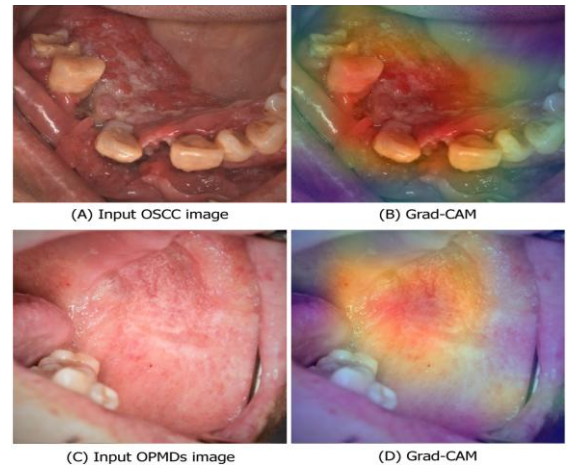


Figure 2: Grad-CAM Visualization

Example Grad-CAM visualizations for the EfficientNet-B0 model. (A) Original photograph with OSCC lesion. (B) Grad-CAM heatmap (overlaid in yellow/red) for the "malignant" class. (C) Original benign case. (D) Corresponding Grad-CAM map. Red/yellow areas indicate regions contributing strongly to the network's decision. The model appropriately focuses on the lesion site in (B) while producing no high-activation region in (D).

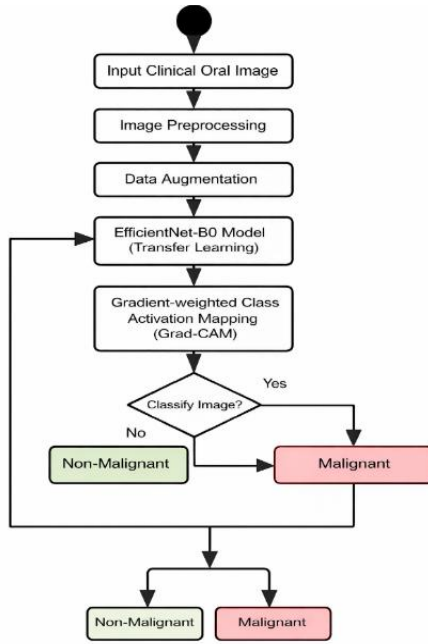


Figure 3: Flowchart

The following flowchart shows the process of an automatic oral cancer screening system with deep learning. It starts by feeding it with input clinical images and then proceeds to preprocessing and data augmentation to enhance image quality and variability. Classification of the processed images is then achieved by having an EfficientNet-B0 model. Grad-CAM emphasizes significant areas that have an effect on the prediction. Lastly, the system is used to categorize the images as either malignant or non-malignant and it helps in early and interpretable diagnosis.

VI. RESULT

A. Baseline Performance (B0)

The EfficientNet-B0 model demonstrated a high level of accuracy in classifying oral malignancies on the basis of clinical images, however, the findings reflect the difference between the controlled training and the actual distribution. The architecture obtained a validation accuracy of 97.4 which indicates high convergence of features during the training stage. But accuracy leveled at about 85 when tested using the independent test set. The generalization gap is a natural result of the large variation in clinical photography, which can include changes in lighting, of anatomical presentation such as focal depth, and even of anatomy itself, and is a reminder that validation metrics are misleading in medical AI.

The discriminative strength of the model is also most accurately seen through its ROC-AUC of 0.997, which suggests that it has almost an outstanding power to distinguish between malignant and non-malign cases. The malignant class also showed a recall (sensitivity) of 85.7% with a precision of 70.6, which created an F1-score of 77.4. Clinically, the most crucial measure is the high recall rate; in the screening scenario, the price of a false negative -a detected cancer-is much more than a false positive. Although the reduced level of accuracy indicates that there is a bias towards false alarming potentially suspicious lesions, this sensitivity bias is directly correlated with our main design goal, which is that the potentially life-

threatening cases should be identified and subjected to additional professional scrutiny.

B. Model Scaling Analysis (B0 vs B1 vs B2)

In order to determine the connection between model capacity and clinical usefulness, we further expanded our comparison to EfficientNet-B1 and B2 models trained in the same conditions. We find that although scaling has a major impact on diagnostic performance, the correlation between the two is highly non-linear.

The variant with the best configuration was EfficientNet-B1, which was once again the best variant given it had almost perfect accuracy (99%) and the false negatives were significantly less than those given by the B0 baseline variant. This surge in performance implies that moderate scaling is an effective method of capturing the intricate, subtle patterns of oral lesions and leaving the generalization required to deal with a wide range of clinical images.

Remarkably, a point of diminishing returns was reached when further scaling to EfficientNet-B2 was performed. Although the B2 model had a greater representational capacity, it had a low accuracy (approximately 97%) and a slight increase in the false negatives relative to B1. This change is probably the beginning of overfitting, when the model gains complexity enough to start learning the noise of the data, instead of its pathology, which is a frequent problem in increasing architectures to finite medical data.

These findings help emphasize a highly significant point about medical AI that making models more complex does not necessarily lead to better performance. Instead, a sweet spot in an architectural design whereby model capacity and generalization are at their most optimal state is present. In this experiment, EfficientNet-B1 provides such a desirable trade-off that it is both highly sensitive and very high-performing and not subject to the flaws of larger models in terms of computational cost and over-parameterization.

C. Error Analysis and Discussion

This sensitivity-based behaviour is validated by a closer look at the confusion matrix. Most of the malignant cases were rightly diagnosed with only 15 false negatives reported. Under screening conditions, these misses must be reduced by all means because early intervention is the best predictor of patient survival. The difference between the high AUC and the test accuracy indicates that although the model is a good way of ranking the probability of the disease, there is still room to further optimize the classification threshold to improve the trade-off between sensitivity and specificity depending on the particular clinical needs.

True label	benign	25	0	4
	carcinoma	0	14	3
	OPMD	3	0	20
		Predicted label		
		benign	carcinoma	OPMD

Figure 4: Confusion Matrix

Table of the result of the test sets classification (rows: true classification, columns predicted classification). The correct prediction of the malignant and benign classes (TP and TN) are in the elements of the diagonal and the incorrect ones are the elements not in the diagonal (FP and FN).

A study of the infrequent false negatives established that the majority of false classifications were in early lesions or lesions that were difficult to see. Even seasoned clinicians find it extremely hard to diagnose at the cases, which indicates that the model faces the same obstacles to its diagnosis as do human experts. The next step towards filling this gap is to ensure that more borderline or early-stage cases are presented in the training pipeline in future versions to fine-tune the appearance of the model.

In the end, these findings prove that our system, which offers an automated first line of defense to oral cancer screening, is an efficient system. The model is appropriate in the early detection case since it has a good balance between the ability to discriminate and the sensitivity. In the future, the maximization of datasets and optimization of thresholds will be crucial to the implementation of this technology as more than just a successful pilot project and a practical, clinical tool.

VII. CONCLUSION

The paper confirms the practicality of the implementation of deep learning, namely, a simplified EfficientNet-B0 model, as a strong initial step to protect against oral cancer screening. We have shown that an automated system can give results with a test accuracy of 85% at the high sensitivity needed in life saving diagnostics by training on a multi-source database of 2,314 clinical images. We find this by analyzing the confusion matrix that the system correctly recognizes the overwhelming majority of malignancies with the false negatives kept to the minimum. With the high-stakes environment of oncology, this performance metric of sensitivity-first is the most essential threshold of any tool that is planned to assist in early intervention.

One of the most important contributions of this work is that it stands the test of time in the face of the noises of a real-life clinical photography. Although lighting and angles variation that is inherent to the intraoral imaging was inevitable, the model still had a remarkable ROC-AUC of 0.997. Moreover, we have not only passed out of a black box approach by also incorporating Grad-CAM visualization. Showing that the predictions of the model are based on clinically relevant lesion features, we give a required level of interpretability, which is critical to achieving clinician trust and integrating AI into the current diagnostics routine.

In comparison to the conventional multi-stage pipelines, which demand the intensive manual preprocessing, our suggested framework provides an alternative frame that is simplified and scalable. It is tailored to specifically deal with the inaccuracy of in-the-field data without losing diagnostic insight. But we know this is pre-emptive step, and not a solution. Even though our findings are encouraging, the fact that our current dataset is rather small is a limitation. We negotiated that even visually subtle lesions or those lesions that are hardest to discern by human eye still pose a challenge to the model, which would suggest that specific consideration should be made of the inclusion of the so-called borderline cases in the subsequent training.

This work highlights an essential idea regarding medical imaging the requirement of principled model choice as opposed to uncritical architectural multifacetedness. We conduct a comparative study of the EfficientNet variants (B0, B1, and B2) and find that a linear relationship does not exist between the larger the representative capacity of a model and its clinical utility. Although EfficientNet-B1 became the most efficient configuration, falsely negative decreased dramatically and accuracy and robustness were strengthened, additional scaling to EfficientNet-B2 initiated a decreasing dimension of returns.

The marginal decrease in performance in the B2 variant indicates the development of overfitting, in which the model is becoming complex enough to start modelling the noise present in the data, instead of the underlying pathology. These results indicate that in order to be able to clinically deploy a model, it is preferable to find the architectural sweet spot where the capacity of the model and generalization are harmonized.

In the future, we will aim to increase the size of our dataset to cover a larger variety of pathologies and patient groups, as well as to consider the issue of multi-class classification to offer more diagnostic details. We will also explore optimization of threshold in order to be able to optimize sensitivity and specificity further. Finally, we aim at the transition and translation of a research model to a point-of-care application where this technology can be implemented on mobile technologies in resource-strained environments. With the combination of these smart screening solutions with the mainstream practice, we hope that we can help close the vast difference in early detection and, above all, the long-term patient outcomes.

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